

CHUTE-DES-PASSES, QC, Canada

WMO: 715220

Lat: 49.840N

Lon: 71.168W

Elev: 398

StdP: 96.63

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.4	(c) -29.7	(d) -36.2	(e) 0.1	(f) -32.2	(g) -33.4	(h) 0.2	(i) -29.5	(j) 9.4	(k) -16.8	(l) 8.5	(m) -16.1	(n) 1.9	(o) 320	(p) 0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	27.0	18.2	25.2	17.2	23.4	16.6	20.4	23.9	19.3	22.6	18.2	21.4	3.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.8	21.7	18.1	13.7	20.6	17.0	12.7	19.5	60.6	23.8	56.7	22.6	53.0	21.5	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.9	(b) 6.8	(c) 5.9	(e) -35.8	(f) 30.3	(g) 2.8	(h) 1.7	(i) -37.8	(j) 31.5	(k) -39.5	(l) 32.5	(m) -41.0	(n) 33.5	(o) -43.1	(p) 34.7		
			DB	WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.0	-17.2	-15.2	-8.5	-0.1	8.0	13.9	16.3	15.2	10.7	3.8	-3.5	-12.0		
	DBStd	12.83	7.48	7.20	7.20	5.16	4.51	4.05	2.90	3.10	4.34	4.15	5.13	6.85		
	HDD10.0	3859	842	707	573	303	93	11	0	2	42	198	406	683		
	HDD18.3	6348	1101	940	831	553	320	142	76	104	232	451	656	941		
	CDD10.0	588	0	0	0	1	32	127	196	164	63	5	0	0		
	CDD18.3	36	0	0	0	0	1	9	14	9	3	0	0	0		
	CDH23.3	391	0	0	0	0	30	126	138	78	19	0	0	0		
	CDH26.7	61	0	0	0	0	6	25	19	9	2	0	0	0		
(14) Wind	WSAvg	2.5	2.4	2.6	2.8	3.0	2.7	2.5	2.2	2.1	2.2	2.4	2.5	2.3		
(15) Precipitation	PrecAvg	1074	67	57	59	75	86	105	128	111	113	104	87	81		
	PrecMax	1233	133	104	102	122	149	165	292	245	226	165	127	127		
	PrecMin	929	20	26	28	29	34	55	58	56	67	41	31	40		
	PrecStd	76	22	19	21	25	28	27	43	38	30	30	20	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.5	3.5	10.8	15.7	26.7	29.7	28.9	28.0	25.7	18.9	10.7	2.7		
		MCWB	3.3	1.9	5.7	7.7	15.0	18.9	19.4	19.2	19.7	14.2	9.5	2.3		
	2%	DB	0.5	0.8	5.9	12.7	22.9	26.7	26.7	25.7	22.6	15.0	7.3	1.0		
		MCWB	0.3	-0.3	2.2	6.3	13.3	17.0	18.1	18.7	18.1	11.7	6.3	0.6		
	5%	DB	-3.4	-1.5	3.4	9.8	19.8	24.5	24.8	23.5	20.2	12.5	4.8	-0.3		
		MCWB	-3.9	-2.5	0.9	4.8	11.8	15.8	17.4	17.6	16.2	10.3	3.9	-0.6		
	10%	DB	-6.0	-3.9	1.5	7.2	16.8	22.0	22.9	21.5	18.0	10.3	2.9	-2.4		
		MCWB	-6.5	-4.8	-0.2	3.0	10.7	14.9	16.8	16.6	15.1	8.6	2.3	-2.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	2.6	6.2	9.5	17.9	21.3	21.7	21.6	21.2	15.2	9.8	2.4		
		MCDB	3.4	2.8	7.8	12.7	21.7	26.2	25.6	25.0	24.5	17.9	10.4	2.6		
	2%	WB	0.3	-0.3	3.0	6.9	14.8	19.4	20.1	20.0	19.1	12.7	6.7	0.7		
		MCDB	0.4	0.7	4.5	11.3	19.3	23.1	23.4	23.5	21.5	14.5	7.2	0.9		
	5%	WB	-3.8	-2.6	1.4	5.3	13.0	17.9	19.0	18.9	17.3	10.7	4.0	-0.7		
		MCDB	-3.6	-1.7	3.0	8.9	18.0	21.7	22.3	22.2	19.6	11.9	4.6	-0.4		
	10%	WB	-6.5	-4.8	0.0	3.8	11.4	16.2	17.9	17.6	15.6	8.8	2.2	-2.7		
		MCDB	-6.0	-4.0	1.4	6.6	16.0	20.3	21.4	20.3	17.4	10.0	2.8	-2.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	11.2	11.6	10.3	11.6	11.9	10.8	10.4	10.0	7.2	6.2	8.1		
		MCDBR	10.8	9.5	11.6	13.9	17.1	15.7	14.1	13.6	13.2	11.4	8.3	7.8		
	5% WB	MCWBR	10.9	8.9	8.5	8.6	9.2	7.8	6.9	7.3	8.5	8.4	7.5	7.8		
		MCWBR	10.9	9.6	9.8	12.1	14.1	12.2	10.8	11.1	11.3	9.9	7.6	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.263	0.290	0.338	0.340	0.371	0.380	0.366	0.352	0.312	0.289	0.245		
		taud	2.294	2.267	2.237	2.256	2.446	2.398	2.391	2.435	2.450	2.547	2.452	2.347		
	Ebn at Noon		812	888	919	905	918	887	872	869	845	830	759	767		
		Edh at Noon	67	89	111	123	107	113	113	103	91	69	59	55		
(44) All-Sky Solar Radiation	RadAvg		1.19	2.25	3.53	4.62	5.28	5.50	5.02	4.14	3.12	1.79	0.94	0.71		
	RadStd		0.17	0.24	0.35	0.37	0.45	0.42	0.45	0.25	0.24	0.18	0.09	0.08		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page